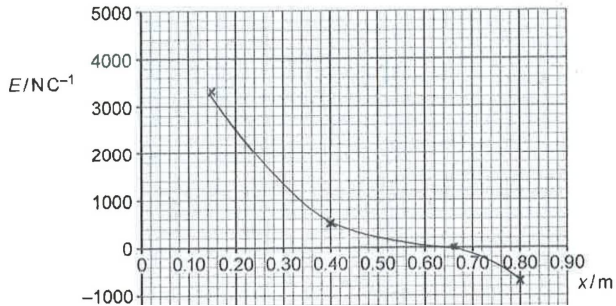


	$= 126 \text{ rad s}^{-1}$
3(a)(ii)	The difference between the normal contact force and its weight provides the centripetal force for circular motion. $F - W = mr\omega^2$ $F - 1.20 \times 9.81 = 1.20 \times 0.230 \times 125.66^2$ $F = 4369.9 = 4370 \text{ N}$ $F = \underline{4370 \text{ N}}$
3(a)(iii)	The contact force by the towel on the drums of the washing machine is large and so when the drum rotates, it will experience large forces at all directions at all positions, causing it to vibrate vigorously with high amplitude. Placing a weight at the bottom of the washing machine will then act as a mass dampener to reduce the amplitude; its natural frequency will also be reduced so that it will not resonate easily; its centre of gravity will be lowered to make it more stable.
3(b)	When spinning at higher angular velocity, a larger resultant force is required to provide centripetal force for the water droplets to move in circular motion. At the position of the hole, there is no contact force and hence the water droplets will not move in circular motion and hence to move outwards, according to Newton's first law.
4(a)	A positive charge produces a positive electric potential. A negative charge produces a negative electric potential. Since the electric potential between A and B, which is a scalar addition of electric potential due to both A and B, and it is positive throughout, both sphere are positively charged. Hence the spheres must have charges of the same sign. The sign of the charge on sphere A is positive.
4(b)	

No.	Solution
	$E = -\frac{dv}{dx} = -\text{gradient of } v - x \text{ graph}$ At $x = 0.15 \text{ m}$, $\frac{dv}{dx} = \frac{700 - 100}{0.12 - 0.30} = -3333.3 \Rightarrow E = 3330 \text{ N C}^{-1}$ At $x = 0.40 \text{ m}$, $\frac{dv}{dx} = \frac{440 - 20}{0.06 - 0.78} = -583.3 \Rightarrow E = 583 \text{ N C}^{-1}$ At $x = 0.80 \text{ m}$, $\frac{dv}{dx} = \frac{280 - 0}{0.90 - 0.50} = 700 \Rightarrow E = 700 \text{ N C}^{-1}$ 
4(c)(i)	$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$ $= \frac{1}{4\pi \times 8.85 \times 10^{-12}} \times \frac{1.0 \times 10^{-8}}{0.15}$ $= 599.45 = 600 \text{ V}$
4(c)(ii)	Fig 2 shows the scalar sum of electric potential due to both A and B. Since A and B